

Bugs found: 4

1. Array index out of bounds: arr has size 3 (valid indices 0-2), but arr[3] accesses index 3.
2. NullPointerException: arr[3] is null by default; calling .length() on null throws NullPointerException.
3. Integer overflow: '1 << 31' shifts a 32-bit int to produce Integer.MIN_VALUE (-2147483648), which is almost certainly unintended.
4. Broken equals/hashCode contract: Bad overrides equals() but not hashCode(), causing incorrect behavior in HashSet (duplicate elements may be inserted).